

Question 1:

Use Cramer's Rule to solve the following set of equations:

$$2x_2 + 5x_3 = 9$$

$$2x_1 + x_2 + x_3 = 9$$

$$3x_1 + x_2 = 10$$

Find the roots then substitute the values back into the equations to check your results.

Solution:

A set of linear equations can be represented in matrix form by:

$$[A]\{X\} = \{B\}$$

[A] is the coefficient matrix:

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 0 & 2 & 5 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \end{bmatrix}$$

{X} is the solution vector:

$$\{X\} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \text{unknowns}$$

And {B} is a column vector of the RHS constants:

$$\{B\} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix} = \begin{Bmatrix} 9 \\ 9 \\ 10 \end{Bmatrix}$$

Cramer's rule states that each unknown in a system of linear equations may be expressed as a fraction of two determinants with denominator D and with the numerator obtained from D by replacing the column of coefficients of the unknown by the constants $b_1, b_2, b_3, \dots, b_n$

$$D = \text{Determinant of } [A] = |A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} 0 & 2 & 5 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \end{vmatrix}$$

For a 3x3 matrix, we can find the determinant using:

$$D = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= 0 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 2 & 1 \\ 3 & 0 \end{vmatrix} + 5 \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} = 0 - 2(0 - 3) + 5(2 - 3) = 0 + 6 - 5 = 1$$

$$D = 1$$

MATLAB code to find the determinant:

```
>> A= [0 2 5; 2 1 1; 3 1 0]; % Defines the [A] matrix
>> det(A) % Finds the determinant of [A]
```

To find the first unknown x_1 :

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}}{D} = \frac{\begin{vmatrix} 9 & 2 & 5 \\ 9 & 1 & 1 \\ 10 & 1 & 0 \end{vmatrix}}{1}$$

$$\begin{vmatrix} 9 & 2 & 5 \\ 9 & 1 & 1 \\ 10 & 1 & 0 \end{vmatrix} = 9 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 9 & 1 \\ 10 & 0 \end{vmatrix} + 5 \begin{vmatrix} 9 & 1 \\ 10 & 1 \end{vmatrix} = 9(0 - 1) - 2(0 - 10) + 5(9 - 10) = 6$$

$$x_1 = \frac{6}{1} = 6$$

Similarly,

$$x_2 = \frac{\begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}}{D}$$

$$= \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix} = \begin{vmatrix} 0 & 9 & 5 \\ 2 & 9 & 1 \\ 3 & 10 & 0 \end{vmatrix} = -8$$

$$x_2 = \frac{-8}{1} = -8$$

MATLAB code to substitute second column in [A] with a column vector {B} and find the determinant:

```
>> A(:,2) = B % Overwrites the second column of [A] with {B}
>> det(A) % Finds the determinant of [A]
```

$$x_3 = \frac{\begin{vmatrix} a_{11} & a_{12} & b_{13} \\ a_{21} & a_{22} & b_{23} \\ a_{31} & a_{32} & b_{33} \end{vmatrix}}{D} = \frac{\begin{vmatrix} 0 & 2 & 9 \\ 2 & 1 & 9 \\ 3 & 1 & 10 \end{vmatrix}}{D} = 5$$

Substitute the values of x_1 , x_2 , and x_3 back into the set of linear equations:

$$2(-8) + 5(5) = 9 \quad \checkmark$$

$$2(6) + (-8) + 5 = 9 \quad \checkmark$$

$$3(6) + (-8) = 10 \quad \checkmark$$

For more than three equations, Cramer's rule is usually impractical, because finding the determinants becomes time-consuming and computationally inefficient.

Question 2:

Use Gauss elimination to solve the following set of equations:

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

Find the roots then substitute the values back into the equations to check your results.

Solution:

$$[A]\{X\} = \{B\}$$

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \\ 6 & 1 & 1 \end{bmatrix}$$

$$\{B\} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix} = \begin{Bmatrix} -2 \\ 4 \\ 6 \end{Bmatrix}$$

$$\begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \\ 6 & 1 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -2 \\ 4 \\ 6 \end{Bmatrix}$$

Apply forward elimination:

$$\text{Row 2} - \frac{a'_{21}}{a'_{11}} \text{Row 1} \rightarrow \text{Row 2}$$

$$\text{Row 2}_{\text{LHS}} = \left\{ 5 - \left(\frac{5}{4} * 4\right) \quad 1 - \left(\frac{5}{4} * 1\right) \quad 2 - \left(\frac{5}{4} * -1\right) \right\} = \{0 \quad -0.25 \quad 3.25\}$$

$$\text{Row 2}_{\text{RHS}} = 4 - \left(\frac{5}{4} * -2\right) = 6.5$$

$$\text{Row 3} - \frac{a'_{31}}{a'_{11}} \text{Row 1} \rightarrow \text{Row 3}$$

$$\text{Row 3}_{\text{LHS}} = \left\{ 6 - \left(\frac{6}{4} * 4\right) \quad 1 - \left(\frac{6}{4} * 1\right) \quad 1 - \left(\frac{6}{4} * -1\right) \right\} = \{0 \quad -0.5 \quad 2.5\}$$

$$\text{Row 3}_{\text{RHS}} = 6 - \left(\frac{6}{4} * -2\right) = 9$$

The resulting system of equations will be:

$$\begin{bmatrix} 4 & 1 & -1 \\ 0 & -0.25 & 3.25 \\ 0 & -0.5 & 2.5 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -2 \\ 6.5 \\ 9 \end{Bmatrix}$$

For the second pivot element ($a'_{22} = -0.25$):

Apply forward elimination:

Row 3 $-\frac{a'_{32}}{a'_{22}}$ Row 2 $\rightarrow$ Row 3

$$\text{Row } 3_{\text{LHS}} = \left\{ 0 - \left(\frac{-0.5}{-0.25} * 0 \right) \quad -0.5 - \left(\frac{-0.5}{-0.25} * -0.25 \right) \quad 2.5 - \left(\frac{-0.5}{-0.25} * 3.25 \right) \right\} = \{0 \quad 0 \quad -4\}$$

$$\text{Row } 3_{\text{RHS}} = 9 - \left(\frac{-0.5}{-0.25} * 6.5 \right) = -4$$

The resulting system of equations forms an upper triangle matrix:

$$\begin{bmatrix} 4 & 1 & -1 \\ 0 & -0.25 & 3.25 \\ 0 & 0 & -4 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -2 \\ 6.5 \\ -4 \end{Bmatrix}$$

Apply back substitution:

$$x_3 = \frac{-4}{-4} = 1$$

$$x_2 = \frac{6.5 - (3.25)}{-0.25} = -13$$

$$x_1 = \frac{-2 - (-13) + (1)}{4} = 3$$

Substitute the values of x_1 , x_2 , and x_3 back into the set of linear equations:

$$4(3) + (-13) - 1 = -2 \quad \checkmark$$

$$5(3) + (-13) + 2 = 4 \quad \checkmark$$

$$6(3) + (-13) + 1 = 6 \quad \checkmark$$

Question 3:

Use Gauss elimination with partial pivoting to solve the following set of equations:

$$2x_1 - 6x_2 - x_3 = -38$$

$$-3x_1 - x_2 + 7x_3 = -34$$

$$-8x_1 + x_2 - 2x_3 = -20$$

Find the roots then substitute the values back into the equations to check your results.

Solution:

$$[A]\{X\} = \{B\}$$

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 2 & -6 & -1 \\ -3 & -1 & 7 \\ -8 & 1 & -2 \end{bmatrix}$$

$$\{B\} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix} = \begin{Bmatrix} -38 \\ -34 \\ -20 \end{Bmatrix}$$

$$\begin{bmatrix} 2 & -6 & -1 \\ -3 & -1 & 7 \\ -8 & 1 & -2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -38 \\ -34 \\ -20 \end{Bmatrix}$$

For the first pivot element ($a_{11} = 2$):

Apply partial pivoting → Determine the largest available coefficient in the column below the pivot element then swap the rows so that the largest element is the pivot element.

$|2| < |-3| < |-8| \rightarrow$ Swap Row 3 with Row 1 so that the new pivot element is -8

$$\begin{bmatrix} -8 & 1 & -2 \\ -3 & -1 & 7 \\ 2 & -6 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -20 \\ -34 \\ -38 \end{Bmatrix}$$

Apply forward elimination:

Row 2 - $\frac{a'_{21}}{a'_{11}}$ Row 1 → Row 2

$$\text{Row } 2_{\text{LHS}} = \left\{ -3 - \left(\frac{-3}{-8} * -8\right) \quad -1 - \left(\frac{-3}{-8} * 1\right) \quad 7 - \left(\frac{-3}{-8} * -2\right) \right\} = \{0 \quad -1.375 \quad 7.75\}$$

$$\text{Row } 2_{\text{RHS}} = -34 - \left(\frac{-3}{-8} * -20\right) = -26.5$$

Row 3 - $\frac{a'_{31}}{a'_{11}}$ Row 1 → Row 3

$$\text{Row } 3_{\text{LHS}} = \left\{ 2 - \left(\frac{2}{-8} * -8\right) \quad -6 - \left(\frac{2}{-8} * 1\right) \quad -1 - \left(\frac{2}{-8} * -2\right) \right\} = \{0 \quad -5.75 \quad -1.5\}$$

$$\text{Row } 3_{\text{RHS}} = -38 - \left(\frac{2}{-8} * -20\right) = -43$$

The resulting system of equations will be:

$$\begin{bmatrix} -8 & 1 & -2 \\ 0 & -1.375 & 7.75 \\ 0 & -5.75 & -1.5 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -20 \\ -26.5 \\ -43 \end{Bmatrix}$$

For the second pivot element ($a'_{22} = -1.375$):

Apply partial pivoting:

$|-1.375| < |-5.75| \rightarrow$ Swap Row 3 with Row 2 so that the new pivot element is -5.75

$$\begin{bmatrix} -8 & 1 & -2 \\ 0 & -5.75 & -1.5 \\ 0 & -1.375 & 7.75 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -20 \\ -43 \\ -26.5 \end{Bmatrix}$$

Apply forward elimination:

Row 3 - $\frac{a'_{32}}{a'_{22}}$ Row 2 $\rightarrow$ Row 3

$$\begin{aligned} \text{Row } 3_{\text{LHS}} &= \left\{ 0 - \left(\frac{-1.375}{-5.75} * 0\right) \quad -1.375 - \left(\frac{-1.375}{-5.75} * -5.75\right) \quad 7.75 - \left(\frac{-1.375}{-5.75} * -1.5\right) \right\} \\ &= \{0 \quad 0 \quad 8.108696\} \end{aligned}$$

$$\text{Row } 3_{\text{RHS}} = -26.5 - \left(\frac{-1.375}{-5.75} * -43\right) = -16.2174$$

The resulting system of equations will be:

$$\begin{bmatrix} -8 & 1 & -2 \\ 0 & -5.75 & -1.5 \\ 0 & 0 & 8.108696 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} -20 \\ -43 \\ -16.2174 \end{Bmatrix}$$

Apply back substitution:

$$\begin{aligned} x_3 &= \frac{-16.2174}{8.108696} = -2 \\ x_2 &= \frac{-43 - (-1.5 * -2)}{-5.75} = 8 \\ x_1 &= \frac{-20 + (2 * -2) - (1 * 8)}{-8} = 4 \end{aligned}$$

Substitute the values of x_1 , x_2 , and x_3 back into the set of linear equations:

$$2(4) - 6(8) - 2(-2) = -38 \quad \checkmark$$

$$-3(4) - (8) + 7(-2) = -34 \quad \checkmark$$

$$-8(4) + 8 - 2(-2) = -20 \quad \checkmark$$

To determine the roots using Gauss elimination method, forward elimination is performed to make [A] an upper triangle matrix, then back substitution is applied to find the unknowns. Partial pivoting is applied to avoid problems that may arise when the pivot element is zero (division by zero) or when the magnitude of the pivot element is smaller than the other elements (round-off errors).